This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the —README.md— for this assignment includes instructions to regenerate this handout with your typeset LATEX solutions.

2.a

The probability of a specific trajectory is the product of the probabilities of the agent choosing an action based on its policy and the environment transitioning to the next state based on that action.

$$\rho^{\pi}(\tau) = p(s_0) \prod_{t=0}^{\infty} \pi(a_t|s_t)\mathcal{P}(s_{t+1}|s_t, a_t)$$

Because we assume that the MDP has a single, fixed starting state s_0, i.e. $p(s_0) = 1$. Therefore,

$$\rho^{\pi}(\tau) = \prod_{t=0}^{\infty} \pi(a_t|s_t)\mathcal{P}(s_{t+1}|s_t, a_t)$$

2.b

Firstly, apply the linearity of expectation and then expand the expectation.

$$\begin{aligned}
\mathbb{E}_{\tau\sim\rho^\pi}\left[\sum_{t=0}^{\infty}\gamma^t f(s_t,a_t)\right] &= \sum_{t=0}^{\infty}\gamma^t \mathbb{E}_{\tau\sim\rho^\pi}[f(s_t,a_t)] \\
&= \sum_{t=0}^{\infty}\gamma^t \sum_{s\in\mathcal{S}}\sum_{a\in\mathcal{A}} p(s_t=s,a_t=a)f(s,a) \\
&= \sum_{t=0}^{\infty}\gamma^t \sum_{s\in\mathcal{S}}\sum_{a\in\mathcal{A}} p(s_t=s)p(a_t=a|s_t=s)f(s,a) \\
&= \sum_{t=0}^{\infty}\gamma^t \sum_{s\in\mathcal{S}}\sum_{a\in\mathcal{A}} p(s_t=s)\pi(a|s)f(s,a) \\
&= \sum_{s\in\mathcal{S}}\sum_{a\in\mathcal{A}} \pi(a|s)f(s,a)\left(\sum_{t=0}^{\infty}\gamma^t p(s_t=s)\right)
\end{aligned}$$

We have the definition of the discounted, stationary state distribution

$$d^\pi(s) = (1-\gamma)\sum_{t=0}^{\infty}\gamma^t p(s_t=s)$$

Rearrange the equation we can get solve $\sum_{t=0}^{\infty}\gamma^t p(s_t=s)$

$$\sum_{t=0}^{\infty}\gamma^t p(s_t=s) = \frac{d^\pi(s)}{1-\gamma}$$

Substitute back and we get

$$\begin{aligned}
\mathbb{E}_{\tau\sim\rho^\pi}\left[\sum_{t=0}^{\infty}\gamma^t f(s_t,a_t)\right] &= \sum_{s\in\mathcal{S}}\sum_{a\in\mathcal{A}} \pi(a|s)f(s,a)\left(\frac{d^\pi(s)}{1-\gamma}\right) \\
&= \frac{1}{1-\gamma}\sum_{s\in\mathcal{S}} d^\pi(s)\left(\sum_{a\in\mathcal{A}}\pi(a|s)f(s,a)\right) \\
&= \frac{1}{1-\gamma}\mathbb{E}_{s\sim d^\pi}[\mathbb{E}_{a\sim\pi(s)}[f(s,a)]]
\end{aligned}$$

2.c

$$
\begin{aligned}
V^{\pi}(s_0) - V^{\pi'}(s_0) &= \mathbb{E}_{\tau \sim \rho^{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t \mathcal{R}(s_t, a_t) \right] - V^{\pi'}(s_0) \\
&= \mathbb{E}_{\tau \sim \rho^{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t \left(\mathcal{R}(s_t, a_t) + V^{\pi'}(s_t) - V^{\pi'}(s_t) \right) \right] - V^{\pi'}(s_0) \\
&= \mathbb{E}_{\tau \sim \rho^{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t \left(\mathcal{R}(s_t, a_t) + \gamma V^{\pi'}(s_{t+1}) - V^{\pi'}(s_t) \right) \right] \\
&= \mathbb{E}_{\tau \sim \rho^{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t \left(Q^{\pi'}(s_t, a_t) - V^{\pi'}(s_t) \right) \right] && \text{Definition of } Q^{\pi'}(s_t, a_t) \\
&= \mathbb{E}_{\tau \sim \rho^{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t A^{\pi'}(s_t, a_t) \right] && \text{Definition of } A^{\pi'}(s_t, a_t) \\
&= \frac{1}{(1-\gamma)} \mathbb{E}_{s \sim d^{\pi}} [\mathbb{E}_{a \sim \pi(s)} [A^{\pi'}(s, a)]] && \text{From 2.b}
\end{aligned}
$$

3.a

1. To follow *Respect for persons*, the researcher must properly inform the students about the study's purpose, the method they will implement including the use of AI chatbot and random assignment and any other possible risk with the student's consent.

2. To minimize possible harms (as stated in *Beneficence*), the researcher should ensure that the students can withdraw from the study at any time without penalty if they feel the chatbot is ineffective.